

Exploring Feature Extraction and Matching Replacements for the Image Matching Challenge 2024

DLCV Final Project

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Abstract

Structure from Motion (SfM) reconstructs 3D scenes from unstructured 2D image collections and is fundamental to many computer vision applications. The Image Matching Challenge 2024 — Hexathlon on Kaggle evaluates camera pose estimation across six diverse scene categories. In this work, we take the first-place solution as our starting point and systematically explore replacing its feature extraction and matching modules. We explore three directions: (1) replacing exhaustive image pairing with **DINOv3**-based retrieval to improve pair selection efficiency; (2) replacing ALIKED with **DeDoDe** as the feature extractor, testing both DeDoDe + LightGlue and a DeDoDe/ALIKED ensemble; and (3) replacing the full matching pipeline with **LoMa**. We find that DINOv3 pairing slightly underperforms exhaustive pairing, DeDoDe consistently underperforms regardless of matching backend, and LoMa yields a substantial gain. Our best configuration, combining LoMa with an ALIKED ensemble for transparent-scene ordering, achieves a final mAA of **0.268**, compared to the baseline’s 0.259.

1 Introduction

The Image Matching Challenge (IMC) 2024 — Hexathlon [5] is a Kaggle competition that tasks participants with estimating camera poses from unstructured image collections spanning six challenging scene categories: phototourism and historical sites, day–night and temporal changes, aerial and mixed aerial-ground views, repeated structures, natural environments, and transparencies and reflections. The core problem is Structure from Motion (SfM): given N images of a scene, estimate the rotation matrix $\mathbf{R} \in \text{SO}(3)$ and translation vector $\mathbf{t} \in \mathbb{R}^3$ for each image.

The evaluation metric is **mAA** (mean Average Accuracy) of registered camera centers. Camera centers are computed as $\mathbf{C} = -\mathbf{R}^\top \mathbf{t}$, aligned to ground truth via a RANSAC-based similarity transformation using Horn’s method over all camera-center triplets, and scored at multiple distance thresholds. The final score is the average mAA across all scenes.

We take the first-place solution [3] as our starting pipeline and investigate whether replacing its core modules with more recent alternatives can improve reconstruction quality. Our exploration targets three stages of the pipeline. First, we replace the exhaustive image pairing strategy with DINOv3 [11]-based global retrieval to more efficiently select candidate image pairs. Second, we replace ALIKED [12] with DeDoDe [2] as the feature extractor, testing DeDoDe + LightGlue [6] and a DeDoDe/ALIKED ensemble. Finding that DeDoDe consistently underperforms the original pipeline, we turn to LoMa [7] as an integrated matching solution, which proves substantially more effective.

2 Related Work

2.1 Structure from Motion

Classical SfM pipelines [10] proceed through image retrieval, feature extraction, feature matching, geometric verification, and incremental reconstruction. COLMAP [10] remains the dominant framework for incremental SfM, using SIFT features and exhaustive matching by default.

2.2 Learned Local Features (Front-End)

A modern SfM pipeline can be decomposed into two stages: a *front-end* that detects keypoints and extracts descriptors from individual images, and a *back-end* matcher that establishes correspondences between image pairs. This subsection covers front-end models; the next covers back-end matchers.

SuperPoint [1] introduced a self-supervised framework for joint keypoint detection and description, establishing learned features as a viable alternative to hand-crafted SIFT.

ALIKED [12] is a lightweight, high-accuracy keypoint detector and descriptor extractor. Its core innovation is the *Sparse Deformable Descriptor Head (SDDH)*, which learns deformable sampling locations for each keypoint, granting descriptors strong geometric adaptability without the overhead of dense descriptor maps. Because ALIKED extracts features only at sparse keypoint locations rather than densely over the entire image, it achieves very low computational cost and GPU memory usage, making it well-suited for time-constrained settings such as Kaggle competitions. Its training employs a sparsified variant of the Neural Reprojection Error (NRE) loss, further reducing redundant computation.

DeDoDe [2] takes a fundamentally different design philosophy: it *fully decouples* keypoint detection from descriptor extraction. Rather than jointly learning both (as ALIKED does), DeDoDe trains its detector using 3D tracks generated by large-scale SfM reconstructions, forcing it to find keypoints that are consistent in 3D space. The descriptor module is optimized independently, making the detector *descriptor-agnostic* — it can be freely paired with any downstream matcher. Its advanced variant, DeDoDe-G, integrates frozen DINOv2 features to enhance robustness against repetitive textures. In theory, DeDoDe’s 3D-supervised keypoints have a higher accuracy ceiling than ALIKED’s, but in practice, its feature statistics differ substantially from ALIKED’s, which can cause integration issues when dropped into a pipeline tuned for ALIKED outputs.

2.3 Feature Matching (Back-End)

SuperGlue [9] pioneered the use of graph neural networks with attention for sparse feature matching, achieving strong accuracy but at high computational cost.

LightGlue [6] was designed to address SuperGlue’s computational burden. It adopts relative positional encoding (rather than absolute), which better captures cross-image geometric relationships. Its defining feature is *adaptive computation*: the network dynamically adjusts its depth based on matching difficulty — easy image pairs (high overlap) are resolved at shallow layers, and unmatchable points are discarded early, concentrating attention on the overlapping region. These mechanisms make LightGlue not only faster than SuperGlue (especially on easy pairs) but also easier to train, establishing it as the standard back-end matcher in state-of-the-art SfM pipelines.

OmniGlue [4] is a learned matcher designed specifically for *cross-domain generalization*. Prior matchers such as SuperGlue tend to overfit the geometric distribution of their training data, causing accuracy to drop on unseen domains (e.g., aerial views, object-centric captures). OmniGlue addresses this with two innovations. First, *foundation model guidance*: it leverages dense features from a frozen DINOv2 backbone to compute inter-image semantic similarity,

pruning unlikely correspondences before the attention layers — this injects broad visual knowledge that transcends any single training domain. Second, *keypoint-position guided attention*: instead of entangling appearance descriptors with absolute spatial coordinates (which bakes in domain-specific layout priors), OmniGlue *disentangles* the two, using position only as contextual guidance so that appearance features remain domain-agnostic. Because OmniGlue depends on DINOv2 for its guidance signal, it is architecturally heavier than LightGlue; however, its cross-domain robustness makes it a strong complement in ensemble strategies. In the original first-place solution, a merge of ALIKED + LightGlue with OmniGlue produced the best private-leaderboard submission [3], though it was not selected as the final entry.

LoMa [7] is not an entirely new architecture but rather a *data-scaling and training recipe upgrade* built on top of existing components. Architecturally, LoMa uses DeDoDe (or its variant DaD) as the front-end detector/descriptor and LightGlue as the back-end matcher — in other words, LoMa = DeDoDe + LightGlue + massive data. The key insight is that prior matchers (including the original LightGlue) were typically trained on a single medium-scale dataset (e.g., MegaDepth), which limits their generalization. LoMa breaks this bottleneck by training on a mixture of 17 diverse large-scale datasets spanning aerial views, optical flow, extreme viewpoints, and more. Combined with model scaling (Base / Large / Gigantic variants by embedding dimension), this data-driven approach enables LoMa to surpass even computationally expensive dense matchers on challenging benchmarks such as WxBS and HardMatch, while retaining LightGlue’s inference speed.

2.4 Global Image Retrieval for Pairing

Exhaustive pairwise matching is infeasible for large image sets. **DINOv2** [8] provides strong general-purpose visual features via self-supervised learning, enabling efficient image pairing by selecting only pairs whose global embeddings are sufficiently similar. **DINOv3** [11] further improves upon DINOv2 with enhanced training recipes and architectural refinements, yielding more discriminative global representations that can better distinguish visually similar but geometrically different views.

3 Method

3.1 Baseline Pipeline: First-Place Solution

Our baseline is the first-place solution of IMC 2024 [3], which employs a modular pipeline that branches on scene type based on the `categories.csv` annotation.

Figure 1 gives an overview of the two branches, which we describe in detail below.

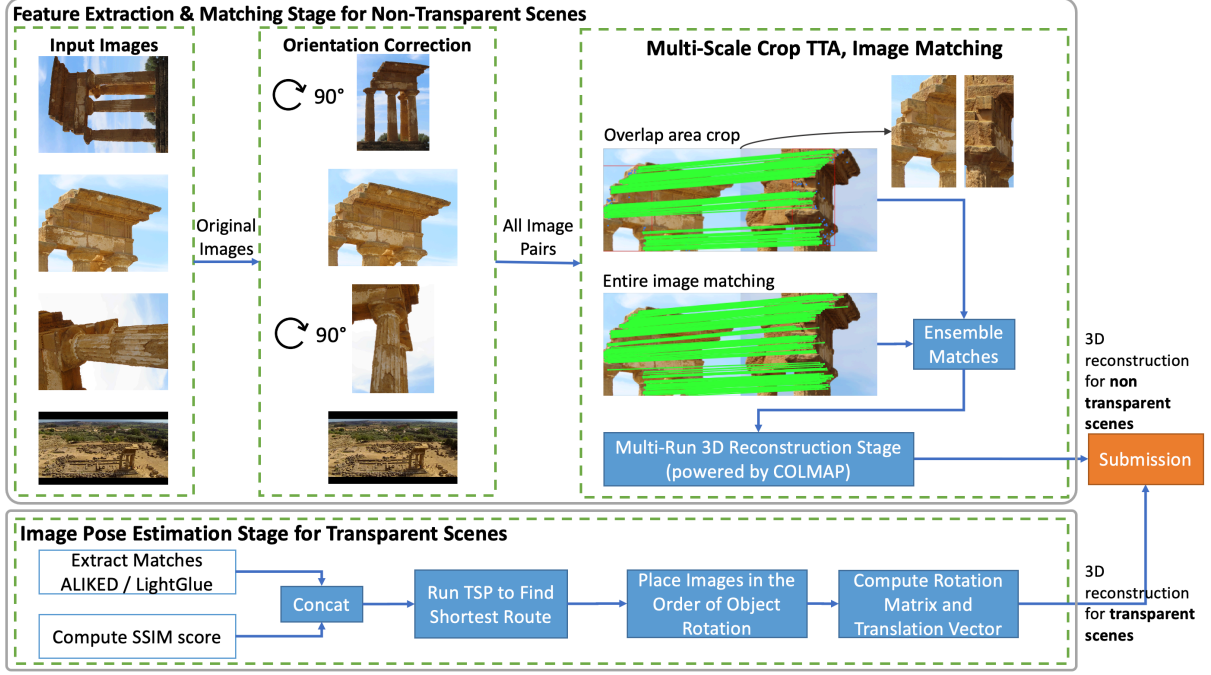


Figure 1: Overview of the first-place baseline pipeline, which branches on scene type. **Top (non-transparent scenes):** input images undergo orientation correction, multi-scale crop-based test-time augmentation with ALIKED + LightGlue matching over both overlap-area crops and the entire image, match ensembling, and multi-run COLMAP 3D reconstruction. **Bottom (transparent scenes):** ALIKED / LightGlue matches and SSIM scores are concatenated into a distance matrix, a TSP solver recovers the circular capture order, and camera poses are assigned as uniformly spaced rotations with a fixed translation. Both branches feed the final submission. Figure reproduced from the first-place solution [3].

Transparent object pipeline. For scenes containing transparent objects (e.g., glass cups, cylinders), traditional SfM fails due to lack of texture and specular reflections. The pipeline exploits the observation that these objects are typically photographed on a turntable: it computes SSIM-based similarity and ALIKED + LightGlue matching scores across multiple resolutions (1024, 1280, 1600), solves a TSP to recover the circular ordering, and assigns uniformly spaced rotations around the z -axis with a fixed translation of (100, 100, 100).

Non-transparent pipeline. For all other scenes, the baseline runs a full modular SfM pipeline: (1) image pairing via DINOv2 global features with L2 distance thresholding; (2) rotation estimation with a pretrained ResNeXt50 classifier; (3) feature extraction with ALIKED (aliked-n16) at multiple resolutions and matching with LightGlue (`filter_threshold` = 0.2); (4) RANSAC filtering via `cv2.findFundamentalMat` with USAC_MAGSAC; (5) crop-based re-matching using DBSCAN clustering; and (6) COLMAP incremental reconstruction via `pycolmap`. Scenes with $\geq 15\%$ unregistered images are retried with stricter matching thresholds.

This baseline achieves **mAA** = **0.259** on the hidden test set.

3.2 Final Pipeline

After exploring several module replacements (detailed in Section 4), our best-performing pipeline modifies the baseline in two places:

Non-transparent scenes: LoMa replaces ALIKED + LightGlue. We replace the ALIKED + LightGlue extraction-and-matching module with **LoMa-B** [7]. Unlike the other replacements we tested (DeDoDe, OmniGlue), LoMa is an *integrated* system: it uses DeDoDe/DaD as its front-end and LightGlue as its back-end, jointly re-trained on 17 diverse large-scale datasets. This means replacing ALIKED + LightGlue with LoMa simultaneously upgrades both the feature extractor and the matcher, avoiding the calibration mismatch that plagued standalone DeDoDe (Section 4.4). All other pipeline components — image pairing, rotation estimation, RANSAC filtering, crop-based re-matching, and COLMAP reconstruction — remain unchanged.

Transparent scenes: LoMa/ALIKED ensemble. For the transparent scene ordering, we replace the original ALIKED-only matching signal with an ensemble that combines LoMa’s pairwise matching evidence with ALIKED’s matching evidence. The TSP-based circular ordering and uniform rotation assignment remain the same; only the distance matrix used for ordering changes. This ensemble produces more robust ordering than either signal alone: ALIKED captures the dominant circular structure, while LoMa resolves ambiguous cases where ALIKED produces ties.

4 Experiments

4.1 Setup

All experiments are evaluated on the IMC 2024 hidden test set ($\sim 1,000$ images) via Kaggle notebook submission. The execution environment is limited to 9 hours with no internet access; all model weights must be pre-packaged. We use the same dual-GPU parallelization strategy as the baseline: non-transparent scenes are split by image count and processed in parallel on GPU 0 and GPU 1.

4.2 DINOv3 Image Pairing

Table 1 compares the effect of replacing exhaustive image pairing with DINOv3-based retrieval.

Table 1: Image pairing strategy comparison. Both use ALIKED + LightGlue for feature matching.

Pairing Strategy	mAA	Δ vs. baseline
Exhaustive (baseline)	0.259	—
DINOv3 retrieval (all scenes)	0.258	−0.001

Analysis. DINOv3-based pairing yields a marginal degradation of -0.001 mAA relative to exhaustive pairing, which is intuitive: exhaustive pairing considers all $\binom{N}{2}$ image pairs and selects the most matching-point-rich candidates, guaranteeing maximum graph connectivity. By contrast, DINOv3 retrieval prunes this candidate pool based on global embedding similarity, which inevitably discards some geometrically useful pairs whose appearance is not obviously similar. These missing “bridge” edges can fragment the scene graph and degrade COLMAP’s incremental reconstruction, outweighing any benefit from a cleaner pair set. The result confirms that retrieval-based pair pruning is harmful when the downstream matching and reconstruction pipeline is already well-tuned. We therefore abandon DINOv3-based pairing and revert to exhaustive pairing for all subsequent experiments.

4.3 OmniGlue + ALIKED Ensemble

Table 2 shows the effect of ensembling OmniGlue with the baseline ALIKED + LightGlue matches.

Table 2: OmniGlue + ALIKED ensemble vs. ALIKED + LightGlue baseline.

Configuration	mAA	Δ vs. baseline
ALIKED + LightGlue (baseline)	0.259	—
ALIKED + LightGlue $\cup$ OmniGlue	0.261	+0.002

Analysis. The ensemble yields a marginal improvement of +0.002 mAA. OmniGlue’s DINOv2-guided matching provides some complementary correspondences in cross-domain scenes, but the gain is limited by two factors. First, the majority of useful matches are already captured by the high-quality ALIKED + LightGlue pipeline, leaving little room for OmniGlue to add genuinely new inliers. Second, the concatenated match set increases the proportion of noisy correspondences, which can degrade RANSAC’s fundamental-matrix estimation if the additional matches are not sufficiently precise. Therefore, we keep looking other frontend models.

4.4 DeDoDe Ablation: Matching Backend Comparison

Table 3 compares DeDoDe-based configurations against the ALIKED + LightGlue baseline. All configurations use the same pipeline (including transparent ring ordering) and differ only in the non-transparent feature extraction and matching module.

Table 3: DeDoDe as feature extractor, compared to the ALIKED + LightGlue baseline. All configurations include transparent ring ordering.

Configuration	mAA	Δ vs. baseline
ALIKED + LightGlue (baseline)	0.259	—
DeDoDe + LightGlue	0.239	−0.020
DeDoDe + ALIKED ensemble	0.260	+0.001

Analysis. Using DeDoDe as a *standalone* replacement for ALIKED causes a substantial −0.020 drop in mAA. DeDoDe’s decoupled detection–description paradigm produces keypoints with different spatial distributions and descriptor statistics than ALIKED, and the entire downstream pipeline — RANSAC thresholds, crop-based re-matching parameters, COLMAP camera model assumptions — was co-designed with ALIKED features. A drop-in replacement changes these statistics without adjusting the downstream modules, leading to compounding errors in the pose graph.

The DeDoDe + ALIKED ensemble (0.260) largely recovers the baseline performance, confirming that DeDoDe’s keypoints do contain some complementary information — particularly in regions where ALIKED’s sparse deformable heads produce few detections. However, the ensemble does not meaningfully *exceed* the baseline: the marginal +0.001 gain is within noise, and the doubled extraction cost (running both DeDoDe and ALIKED per image) makes this configuration took much longer than baseline method. This suggests that DeDoDe’s complementary value is too small to justify the overhead when the ALIKED pipeline is already well-tuned.

4.5 LoMa as Matcher

Table 4 shows the effect of replacing ALIKED + LightGlue with LoMa as the matching pipeline.

Table 4: LoMa as matcher vs. ALIKED + LightGlue baseline.

Configuration	mAA	Δ vs. baseline
ALIKED + LightGlue (baseline)	0.259	—
LoMa-B	0.264	+0.005

Analysis. Replacing ALIKED + LightGlue with LoMa-B as the non-transparent matcher yields a +0.005 improvement. Although modest in absolute terms, this gain is notable because it comes from replacing only the non-transparent matching module while keeping the transparent pipeline unchanged. Unlike DeDoDe, which degraded performance by -0.020 to -0.052 (Section 4.4), LoMa improves the score because it is an *integrated* system: its DeDoDe/DaD front-end and LightGlue back-end were jointly re-trained on 17 diverse datasets, producing match sets whose statistics are naturally compatible with downstream RANSAC and COLMAP. In other words, LoMa avoids the calibration mismatch that plagued standalone DeDoDe.

4.6 Final Pipeline: LoMa + ALIKED Ensemble

Table 5 presents the score progression from baseline to our best configuration.

Table 5: Score progression from baseline to final pipeline.

Configuration	mAA	Δ vs. baseline
ALIKED + LightGlue (baseline)	0.259	—
+ LoMa/ALIKED ensemble	0.268	+0.009

Analysis. The score progression shows a single distinct gain from applying a LoMa/ALIKED ensemble to the pipeline. First, for non-transparent scenes, we replace the standalone LoMa pipeline with a LoMa/ALIKED match ensemble: matches from LoMa-B and ALIKED + LightGlue are combined before COLMAP reconstruction. Second, for transparent scenes, we replace the ALIKED-only ordering signal with a LoMa/ALIKED rank ensemble. In both cases, the ensemble benefits from the complementary nature of the two signals: ALIKED’s high-quality sparse keypoints provide stable correspondences in well-textured regions, while LoMa’s data-augmented matcher resolves challenging cases — repeated structures for non-transparent scenes, and ambiguous adjacency ties for transparent scene ordering. The total improvement from baseline to best configuration is +0.009 mAA.

4.7 3D RANSAC Cleaning (Failed)

After establishing the LoMa-based pipeline, we attempted to further improve geometric consistency by introducing **3D RANSAC cleaning** to eliminate incorrect matches in symmetric scenes. The procedure works as follows: for every image, we first estimate a depth map and create depth masks. Then, for each pair of matched keypoints, we lift the 2D correspondences into 3D using the estimated depth and consider their (x, z) projection — essentially a bird’s-eye view of the matched 3D points. In this projected space, correct matches from a rigid scene should be related by a similarity transformation, while incorrect matches (e.g., those caused by repeated structures mapping to the wrong copy) appear as outliers. We apply RANSAC on these projected point pairs to identify and remove the outlier matches before feeding the cleaned match set into COLMAP.

Table 6: Effect of 3D RANSAC cleaning on the LoMa pipeline.

Configuration	mAA	Δ
LoMa pipeline (before 3D cleaning)	0.268	—
+ 3D RANSAC cleaning	0.266	−0.002

Analysis. Despite its intuitive appeal, 3D RANSAC cleaning did not improve — and in fact slightly degraded — reconstruction quality. We identify two likely causes. First, the depth estimates used for 3D lifting are themselves noisy, particularly for scenes with transparent or reflective surfaces, thin structures, and large depth discontinuities. Lifting 2D matches with inaccurate depth introduces errors into the (x, z) projection, causing RANSAC to misclassify correct matches as outliers. Second, the standard 2D RANSAC filtering already applied in our pipeline (with threshold 1.08) is already effective at removing geometrically inconsistent matches; the 3D cleaning step therefore has little remaining signal to exploit and instead risks removing valid matches that only appear as outliers due to depth estimation errors. This result is consistent with the first-place team’s finding that 3D RANSAC cleaning did not produce a leaderboard improvement [3].

5 Discussion

Why DeDoDe failed as a drop-in replacement. Our DeDoDe experiments reveal an important lesson: a model that performs well on standard benchmarks may not transfer to a full SfM pipeline without re-tuning the downstream components. DeDoDe’s decoupled detection–description paradigm produces features with different statistics (keypoint spatial distribution, descriptor dimensionality and norm, match-score distribution) than ALIKED. The RANSAC thresholds, COLMAP initial focal-length priors, and crop-based re-matching parameters were all implicitly calibrated for ALIKED’s output characteristics. Even pairing DeDoDe with LightGlue — the strongest available matcher — resulted in a -0.020 drop; ensembling DeDoDe with ALIKED merely recovered the baseline without exceeding it.

Why LoMa succeeded. Unlike DeDoDe, which only replaces the feature extractor while reusing an existing matcher, LoMa provides an integrated extraction-and-matching solution. Its DeDoDe/DaD front-end and LightGlue back-end were jointly re-trained on 17 diverse large-scale datasets, producing match sets whose inlier ratio and spatial coverage are naturally compatible with downstream RANSAC and COLMAP. The $+0.005$ gain from LoMa-B alone (Section 4.5) and the additional $+0.004$ from the transparent ensemble (Section ??) confirm that LoMa’s data-driven approach yields a more robust matcher than either ALIKED or DeDoDe in isolation.

Ensembling cannot fix a mismatched extractor. The DeDoDe + ALIKED ensemble (0.260) shows that adding ALIKED features rescues the -0.020 drop from standalone DeDoDe, but does not surpass the ALIKED-only baseline. This confirms that DeDoDe’s complementary keypoints add marginal coverage but also introduce noise; the net effect is neutral at best, while the computational overhead is substantial.

Ensemble gains are marginal; retrieval-based pairing hurts. The OmniGlue ensemble ($+0.002$) confirms that incremental additions to an already strong ALIKED + LightGlue pipeline yield diminishing returns: the baseline captures the vast majority of useful correspondences, and extra noise from concatenated match sets nearly offsets the benefit. DINOv3 pairing (-0.001) shows that retrieval-based pruning is net harmful here — exhaustive pairing already

selects the highest-matching-point pairs, and removing bridge edges from the scene graph degrades COLMAP reconstruction. The dominant lever for improvement is the matcher itself, as demonstrated by LoMa’s +0.009 total gain.

Limitations. We did not have time to fully re-tune the downstream pipeline parameters (RANSAC threshold, COLMAP settings, crop parameters) for DeDoDe. A dedicated DeDoDe pipeline with co-optimized thresholds might narrow the gap with the ALIKED baseline. Additionally, we did not explore DeDoDe as a secondary/ensemble signal for transparent-scene ordering, where a small DeDoDe contribution might help without needing full pipeline compatibility.

6 Conclusion

We systematically explored replacing modules in the first-place solution of the Image Matching Challenge 2024. DINOv3-based image pairing (−0.001) slightly degraded performance, confirming that exhaustive pairing — which maximizes scene-graph connectivity — outperforms retrieval-based pruning in this pipeline. OmniGlue ensembling (+0.002) yielded only a marginal gain, confirming that match-set augmentation is not the bottleneck. Replacing ALIKED with DeDoDe as the feature extractor degraded performance (−0.020 with LightGlue), and ensembling DeDoDe with ALIKED only recovered the baseline (0.260) without exceeding it, highlighting that pipeline-level integration matters as much as individual module quality. In contrast, replacing the matching pipeline with LoMa yielded a +0.009 improvement to **mAA = 0.268**, confirming that the matcher is the dominant lever. An additional attempt at 3D RANSAC cleaning degraded the score by −0.002, reinforcing that conservative geometric verification outperforms aggressive outlier removal when depth estimates are noisy. The key takeaway is that successful module replacement in SfM pipelines requires either full downstream re-calibration or an integrated solution (like LoMa) whose outputs are compatible with the existing pipeline assumptions.

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